SNAKE / INTERACTIVE PCB CARD

Rev B-PUBLIC | 85.6 x 54 mm | USB-C powered | 10 x 9 red LEDs

01 Power USB

5 V input / CC resistors
ESD / LDO / power test pads

File: 01_Power_USB.kicad_sch

02 MCU Inputs Debug

MCU / 4-way + START
SWD / reset / decoupling

File: 02_MCU_Inputs_Debug.kicad_sch

03 LED Driver

I2C address 0x50
10 CS sources / 9 SW scans

File: 03_LED_Driver.kicad_sch

05 NFC

ST25TN01K-AFH5
No USB / MCU power connection

File: 05_NFC.kicad_sch

04 LED Matrix

D1-D90, row-major order
Anode = COL / cathode = ROW

File: 04_LED_Matrix.kicad_sch

INTERCONNECTS

+3V3 and GND: power, MCU and driver
I2C_SCL / I2C_SDA / LED_EN: MCU to driver
LED_COL1..10 / LED_ROW1..9: driver to array
NFC_AC0 / NFC_AC1: isolated passive RF circuit

BUILD NOTES

USB supplies power only. Program through SWD with target powered by USB.
Driver firmware: R_EXT = 20k, GCC = 0x12; unused channels off.
NFC L1 is a printed loop to be designed and tuned during PCB layout. C30 is DNP initially.
No battery, charger, USB data or USB-PD negotiation. QR and appearance belong to PCB artwork.
All circuits are native editable KiCad symbols and wires. Engineering checks accompany exports.

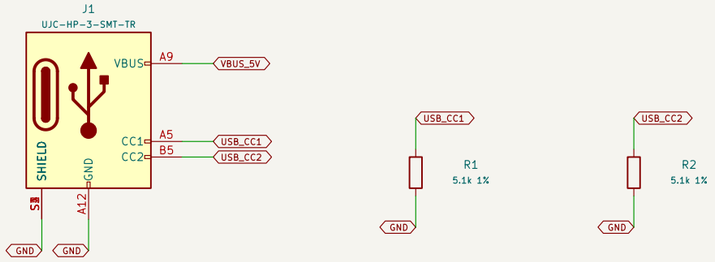
Public reference design — hardware validation pending
USB-C 5V | 10x9 red LED | passive NFC
Snake Interactive PCB Business Card

Sheet: /
File: SnakeBusinessCard.kicad_sch

Title: System overview and implementation notes

Size: A3	Date: 2026-09-15	**Rev: B-PUBLIC**
KiCad E.D.A. 10.0.6		Id: 1/6

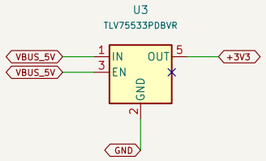
USB-C POWER INPUT

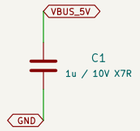


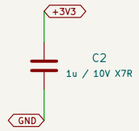
Independent Rd on CC1 and CC2.
Power-only connector: no D+/D- or SBU.
A9/B9 = VBUS; A12/B12 and all shield anchors = GND.

3.3 V SUPPLY





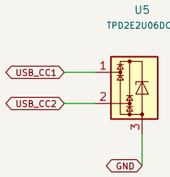






Use regulated 5 V. No 9/12 V PD request.
LDO rated input maximum = 5.5 V.
Normal load target <50 mA; 100 mA budget.
LDO 500 mA capability is NOT a USB current entitlement.
Account for output-capacitor charging at hot plug.

INTERFACE TRANSIENT PROTECTION





Place protection at connector, with short ground return.
TVS is transient protection, not DC overvoltage regulation.
Board-level ESD and hot-plug performance require testing.

POWER ACCESS / MEASUREMENT







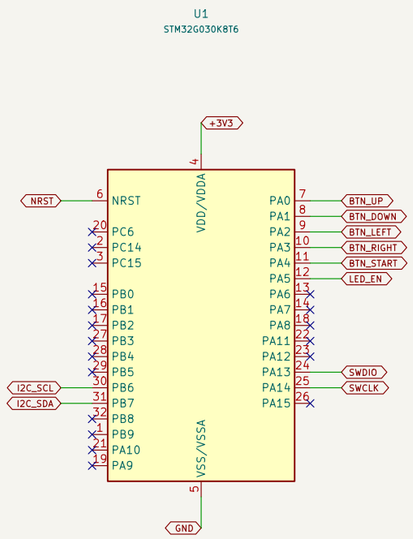
Public reference design — hardware validation pending
USB-C 5V | 10x9 red LED | passive NFC
Snake Interactive PCB Business Card

Sheet: /01 Power USB/
File: 01_Power_USB.kicad_sch

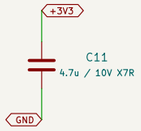
Title: USB-C input, ESD and 3.3 V regulation

Size: A3	Date: 2026-09-15	**Rev: B-PUBLIC**
KiCad E.D.A. 10.0.6		Id: 2/6

MCU / HSI16 INTERNAL CLOCK





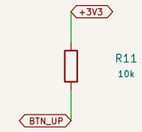




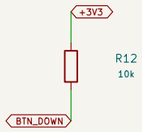


U1 LQFP32: pin 4 includes VDDA; pin 5 includes VSSA.
No external VBAT, VREF+ or exposed pad on this package.
PA13 / PA14 are reserved for SWD. Boot from internal Flash.
Optional UART omitted; PA9/10/11/12 have no-connect marks.

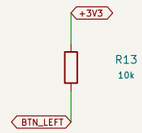
FOUR DIRECTIONS + START





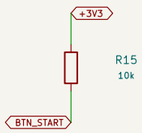














Active low; sample 5 ms / debounce 20 ms in firmware.
START is a game key, not the hardware reset.

SWD CONTACT PADS











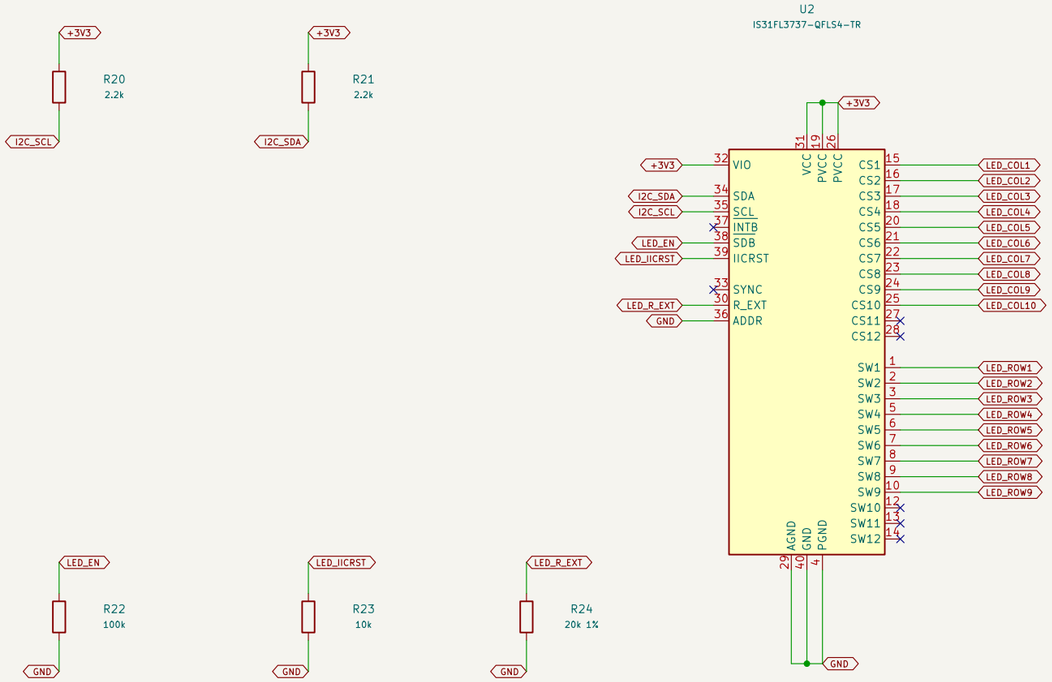
Public reference design — hardware validation pending
USB-C 5V | 10x9 red LED | passive NFC
Snake Interactive PCB Business Card

Sheet: /02 MCU Inputs Debug/
File: 02_MCU_Inputs_Debug.kicad_sch

Title: STM32G030, five controls and SWD

Size: A3	Date: 2026-09-15	**Rev: B-PUBLIC**
KiCad E.D.A. 10.0.6		Id: 3/6

LED DRIVER / 3.3 V LOGIC AND LED POWER

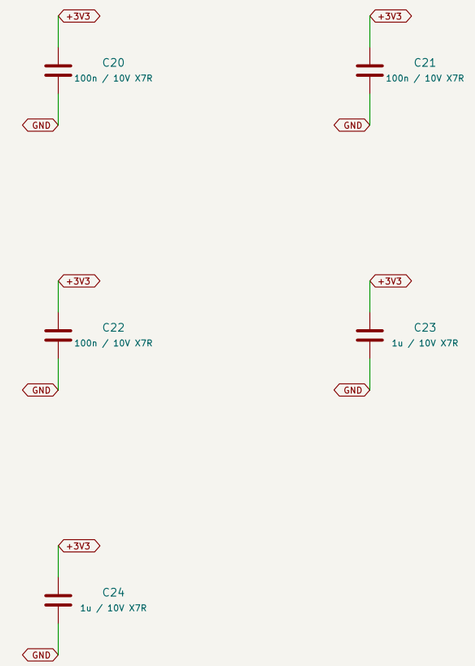


ADDR = GND: 7-bit I2C address 0x50.
IICRST held LOW in normal operation.
SDB (LED_EN) LOW until safe initialization.
SYNC and INTB not used.

FIRMWARE / CURRENT LIMIT

REXT = 20k; GCC = 0x12; highest pixel PWM = 255. About 2.95 mA pulse / 0.231 mA average per LED.
12 scan slots remain active with 9 populated rows. Unused CS11/12 and SW10/11/12 stay disabled.
Hold SDB low; set function page to software shutdown / GCC=0; release SDB; clear pixels; set GCC; enable SSD last.

LOCAL DECOUPLING

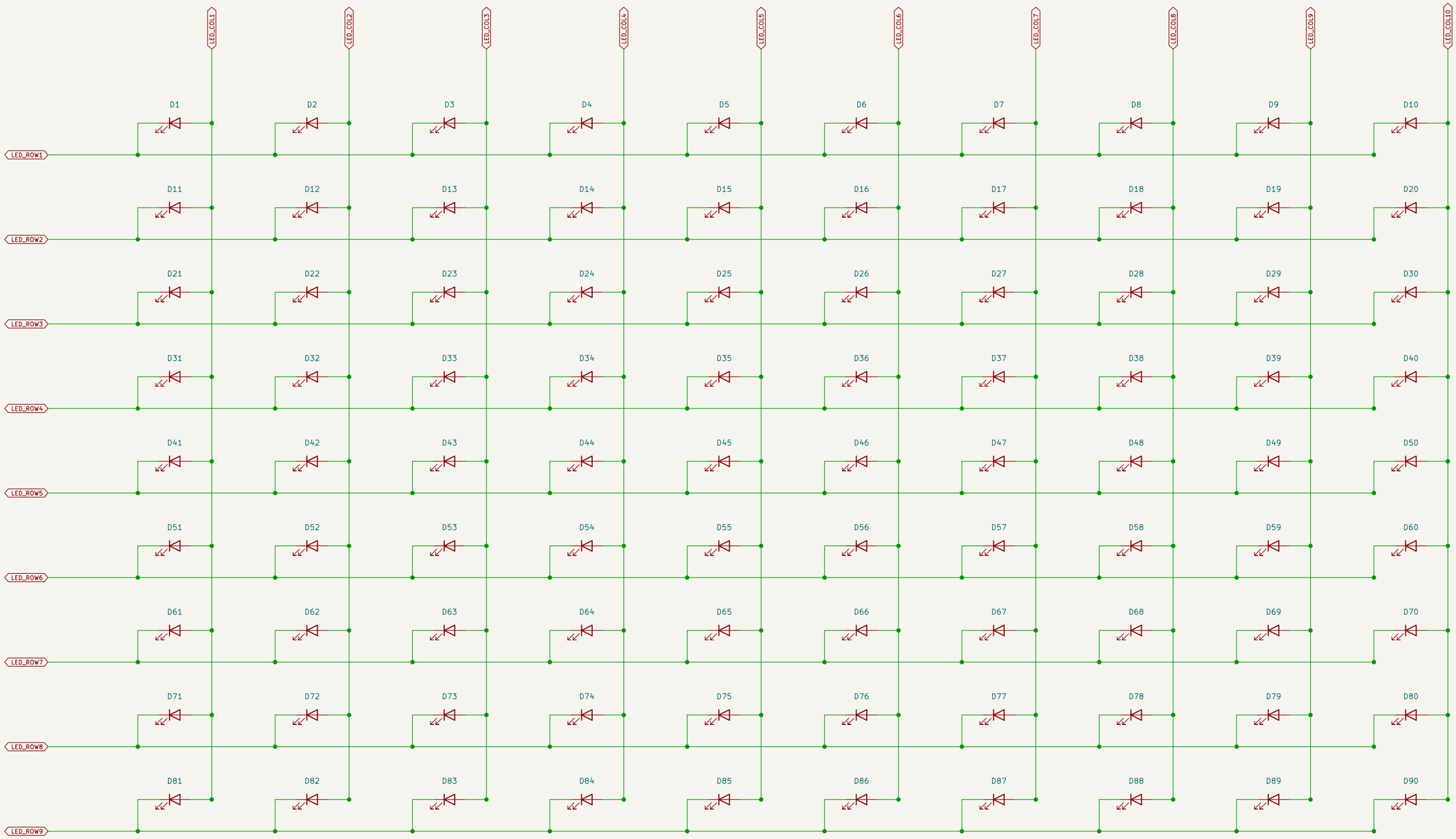


C20: VCC31; C21/C22: PVCC19/26.
C23: PVCC bulk; C24: VIO32.
Keep supply loops short; EP41 to GND.
R_EXT ground away from LED pulse return.

Public reference design — hardware validation pending
USB-C 5V | 10x9 red LED | passive NFC
Snake Interactive PCB Business Card
Sheet: /03 LED Driver/
File: 03_LED_Driver.kicad_sch
Title: IS31FL3737 constant-current matrix driver
Size: A3 | Date: 2026-09-15 | **Rev: B-PUBLIC**
KiCad E.D.A. 10.0.6 | Id: 4/6

90 RED SMD LEDs / APT1608SURCK

x = 0..9 left to right; y = 0..8 top to bottom. D(y*10+x+1). Anode to COL; cathode to ROW.



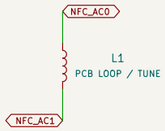
Each CS pin is a constant-current source; no individual series resistors. Use one LED polarity throughout.
This is the electrical connectivity map. The PCB uses a 3.5 mm physical pitch and a compact 10 x 9 grid.

Public reference design — hardware validation pending
USB-C 5V | 10x9 red LED | passive NFC
Snake Interactive PCB Business Card
Sheet: /04 LED Matrix/
File: 04_LED_Matrix.kicad_sch

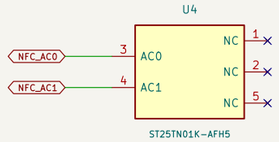
Title: 10 columns x 9 rows — 90 red LEDs

Size: A3	Date: 2026-09-15	**Rev: B-PUBLIC**
KiCad E.D.A. 10.0.6		Id: 5/6

PASSIVE NFC / ST25TN01K-AFH5







Printed antenna, not a purchased inductor.
Initial equivalent-L design target ~2.58 uH.
Final geometry and tuning belong to PCB layout.

Internal tuning capacitance: 50 pF.
AC0 = pin 3; AC1 = pin 4.
Pins 1/2/5 are NC. No VCC or GND.

LAYOUT / VALIDATION

Reserve a separate left-side antenna zone. Keep unrelated copper, ground planes and signal routes out on all layers.
Place the USB connector and switching LED current paths outside this zone. NFC works without USB power.
Tune the assembled tag per ST AN2866, with a coupling loop / VNA and real phones; target resonance 13.6–14.0 MHz.
C30 is not fitted initially; select C0G value only after measurement. Do not treat the illustrative loop value as validated.
Write a short HTTPS NDEF URI after the real portfolio URL is confirmed. QR will point to the same URL.

Public reference design – hardware validation pending
USB-C 5V | 10x9 red LED | passive NFC
Snake Interactive PCB Business Card

Sheet: /05 NFC/
File: 05_NFC.kicad_sch

Title: Independent passive NFC tag and PCB antenna

Size: A3	Date: 2026-09-15	**Rev: B-PUBLIC**
KiCad E.D.A. 10.0.6		Id: 6/6